United International **University (UIU)**

Dept. of Computer Science & Engineering (CSE)

Final Exam: : Trimester: Fall 2025

Course Code: CSE 1111, Course Title: STRUCTURED PROGRAMMING LANGUAGE

Time: 2 hours Total Marks: 40

(Unfair means in the exam will be taken seriously as per UIU disciplinary rules)

Answer all the Questions. Write your serial number of exam attendance sheet on your answer script.

1 a) Write a **program** using **two user defined functions** to perform the following operations 5

i) The function **main()** reads two integer numbers from keyboard and pass the numbers to the **CalValue()** function.

ii) The user defined function **Differ(int x, int y)** checks whether the difference of x and y is positive or negative. If it is positive, it will return **1**, otherwise **0**.

iii) The user defined function **CalValue(int x, int y)** calculates the value of (x+y)(x-y) if the **Differ()** function returns 1, and otherwise calculates the value of (y+x)(y-x). This function returns the calculated value to the calling function.

iv) The **main()** function prints the returned value from the function **CalValue()**.

b) Show **manual tracing** for the following program (below left) and find **output** 5

```
#include<stdio.h>
int a;
float func1(int x);
void func2(float x, int y);

int main(){
    a=11;
    printf(“%f\n”, func1(a+50));
    func2(6.5, 10);
    printf(“%d\n”, a);
    return 0;
 }
 float func1(int x) {
    a=x*10;
    printf(“%d\n”, x+25);
    return x+a+20.5;
 }
void func2(float x, int y){
    printf(“%f\n”, x+5);
    a=a+y;
 }
```

Program for 1 (b)

```
char str1[13]={‘\0’};
char str2[4]={‘\0’};
int i, j, k;

strcpy(str1, “CSE DEPT ”);
strcpy(str2, “UIU”);
i=strlen(str1);
printf(“%s  %s  %d\n”, str1, str2, i);

k=0;
for(j=i; j<i+strlen(str2); j++){
    str1[[j]=str2[k++];
}
printf(“%s  %s  %d\n”, str1, str2, j);

if (strcmp(str2, str1)<0)
   printf(“String %s is greater”, str1);
else
   printf(“String %s is greater”, str2);
```

Program for 2 (a)

2 a) Show **manual tracing** for the program above right. 4

b) Show **manual tracing** for the following program and find **output**. 4

```
#include<stdio.h>

void func(int a, int *b){
   int *q;
   q=b;
   *q=*q+a;
   a=a-3;
}
int main(){
   int a=5, b=9;
   printf("%d  %d\n", a+b, b);
   func(a, &b);
   printf("%d   %d", a+b, b);
   return 0;
}
```

Program for 2 (b)

```
#include<stdio.h>
void func(int B[], int n);
void main(){
   int a[5]={10, 20, 30, 40, 50};
   int i;
   for(i=0; i<5; i++)
      printf("%d  ", --a[i]);
   func(a, 5);
   printf("\n");
   for(i=0; i<5; i++)
      printf("%d  ", ++a[i]);
}
void func(int b[], int n){
   int i;
   for(i=0; i<n; i++){
       b[i]=b[i]%3+1;
   }
}
```

Program for 3 (a)

3 a) Show manual tracing for the program (above right) and find output. 5

b) i) Distinguish between write and append modes in file writing. 2

ii) What type of problems will be happened if a file is not closed after opening? 2

iii) What are the uses and syntax of the following standard functions? 2

fopen(), fprintf(), fscanf(), fclose()

4 Write a **program** having the structure **student (**name, id, course-I marks, course-II marks, phone, email**)** to perform the following operations for two students 7

a) Read name, id, course-I marks, course-II marks, phone and email of two students from keyboard

b) Find the average of the marks of the two courses for each student

c) Display the following **sample** report on monitor for these two students:

```
Rahim    10    80.0   90.0   01632078812   xxx@gmail.com
Saiham   20    70.0   80.0   01710012300   yyy@hotmail.com

Average Marks of Rahim is 85.5
Average Marks of Saiham is 75.5
```

5 Write a **code segment** to reverse an input string without using any string related standard function(s). 4

Input String="CSE Dept"
Output String="tpeD ESC"